G0B75A: Meta-analysis

Master of Statistics
KU Leuven

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Practical information

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Aims of the course

1. To be able to **read** and critically analyse a meta-analytic report,
2. to have **insight** in the strengths and limitations of meta-analysis, and
3. to be able to **conduct** independently a meta-analysis, this is to:
   - formulate a meta-analytic research question or hypothesis,
   - conduct a systematic literature search,
   - extract and code the relevant data,
   - calculate effect sizes based on the reported study outcomes,
   - choose and apply the appropriate statistical models and techniques to analyse the effect sizes, and
   - interpret and report the results of the meta-analysis.

Approach?

- Video lectures: introduction to meta-analysis and to some more advanced topics
- Collective and individual live sessions (discussion, Q&A)
- Independent work: read the obligatory papers, go through the textbook, and (at least for your paper) look for additional literature
- Paper (by preference per two students).

Video lectures

Introduction to meta-analysis

- What is meta-analysis? (video 1)
- Examples and alternative procedures (video 2)

Preparing your dataset

- Study retrieval (video 3)
- Avoiding bias (video 4)
- Data extraction (video 5)
- Effect size calculation (video 6)

Combining effect sizes:

- Meta-analytic models (video 7)
- Assessing the credibility of the meta-analytic results (video 8)
- Meta-analysis with R (video 9)

Video lectures

Meta-analysis of dependent effect sizes

- Multilevel meta-analysis (video 10)
- Robust Variance Estimation (video 11)

Applications and extensions

- SCED meta-analysis (video 12)
- Network MA (video 13)
- MA of diagnostic studies (video 14)
- Meta-analytic SEM analyses (video 15)

Discussing and reporting results (video 16)

Using tools (video 17)

Live meetings

- There are 4 sessions:
  - Session 1 (March 13; on campus with live streaming): practical information + overview of course + discussion and possibility to ask questions regarding the first eight video lectures

  <u>Preparation</u>: look at first eight video lectures. Start reading in the handbook. Mail your questions in advance (by March 11, noon).

  - Session 2 (March 27): exercises with R (one session on campus, one online)

  <u>Preparation</u>: look at the ninth video lecture. Try out the metafor package.

  - Session 3 (April 3): discussion and opportunity to ask questions on lectures and assignment

  <u>Preparation</u>: look at the last eight video lectures. Mail your questions in advance (by April 1)

  - Session 4 (April 24; online individual meeting): opportunity to ask questions on lectures, R-exercises and assignment

Learning material

- Learning material
  - Screencasts and handouts (Toledo)
  - Additional documents, links, references, … (Toledo)
  - Internet and library
  - Textbook:

  Cooper, H. (2017). *Research synthesis and meta-analysis. A step by step approach.* (4th ed.) London: Sage.

  (you can find scans of this book under Course documents)

  Note: we will not closely follow this textbook. The textbook however includes most important information. If you want to know more about specific topics, you are referred to the document with references on Toledo.

Paper

- Paper (more information on Toledo > Assignments):
  - Performing a meta-analysis,
  - critical discussion and re-analysis of existing meta-analysis, or
  - elaborating a methodological/statistical topic.
- Note: simply summarizing a topic from the handbook or the slides, or summarizing an existing meta-analysis is NOT sufficient. An **own contribution** (e.g., bringing together ideas from several sources, or doing own analyses) as well as a **critical discussion** is required!
- If you use statistical software, please add the **syntax** used (e.g., R-script) as an appendix to your paper.

Preparation for exam

- Screencasts (and handouts)
- Handbook Cooper (2017) (see Toledo) or similar introductory handbook.
- Additional texts: Bonett (2008) & Ionnidis (2005)
- Preparing your paper will require searching additional literature! (cfr. Toledo for some suggestions)

Evaluation?

- Paper (10/20 points)
- Oral (10/20 points)
  - Online. 40 minutes of preparation time (with your camera + micro switched on + screen sharing), followed by 15 minutes oral answering (and additional questioning)
  - Open book (you can use your notes, textbook, handouts, ...). You can use these documents on your PC (you do not have to print them).
  - Typing and the use of internet (including AI), e-mail, a cell phone, or other means of communication are NOT allowed.
  - Also includes questions on your paper (so your paper has a very high weight in the evaluation!).

Use of AI

- If you made use of AI for preparing your paper, be transparent about how you used AI. To that end:
  - Fill in the form on AI use that you find on Toledo.
  - Describe in a separate section what prompts you gave, what came out and how you used (or did not use) this output.
  - Include both parts **as two appendices** to your paper.
- Never use AI without insight in the topic. During the oral exam, I will check whether you understand and can explain what you wrote in the paper.
- During the oral exam, you are not allowed to use AI.

Exam

- During the oral exam about 4 questions are asked, including 2 or 3 questions on the paper:
  - Informative questions (e.g., explain a specific formula or sentence from your paper).
  - Questions testing your insight (e.g., can you explain why in your paper you got different results compared to the original meta-analysis?)
- No questions on software will be asked (although output of any software can be given with the question to interpret the results, but even without having used this software, it should be possible to answer this question). If for your paper you used software, please add the syntax in an appendix. During the exam, I can ask for clarifications of the syntax.

Evaluation (continued)

- Kind of questions:
  - **knowledge** is required (e.g., explain the problem of publication bias), but this is not the main focus;
  - **insight** (e.g., a meta-analyst found a negative correlation between effect size and study size. What would be your advice?), and
  - **applications** (e.g., interpret the given output of a meta-analysis).
- See Toledo for example questions (Course Information)
- Note: a lot of topics are discussed in the video lectures, but for some topics (e.g., SCED MA, NMA, diagnostic MA, MASEM) only a first introduction is given. Of course I do not expect that you become an expert in all these topics, but I would like you to get the main ideas for all these topics. If you want to go deeper in a topic, you can do so in your paper.

Evaluation (continued)

- Need help?
  - Ask questions during the four meetings
  - Discussion boards on Toledo
  - Help each other!

Good luck and enjoy!